AI Roundtable

A Real-Time Multi-Agent Podcast with Live Human Interruption

A voice-native system where autonomous AI experts collaborate, debate, and exchange sharp perspectives on any topic—while allowing a human listener to interrupt and steer individual agents in real time.

AUTONOMOUS DEBATE

Role-Based Prompting

Moderator, Optimist, and Skeptic clash with dynamic friction, avoiding boring monologues.

VOICE AI SYNTHESIS

ElevenLabs Turbo v2.5

Low-latency streaming speech with distinct vocal profiles (Rachel, Adam, Nicole).

CORE INNOVATION

Real-Time Interruption

Instant 0ms audio cancellation; context memory updates; agents immediately pivot to listener.

TEAM: Shubham Raj Rishabh Choudhary Priyam Garg

The Hard Part: Beyond Generating Three AI Responses

The challenge isn't simply querying an LLM three times. The hard part is coordinating agents so the podcast sounds coherent, doesn't repeat itself, feels intellectually lively, and reacts immediately to interruptions.

✕ Current AI Voice Limitations

- **Exhaustive Monologues:** Standard LLMs generate 200-word blocks of text that ruin audio podcast cadence.
- **Zero Organic Friction:** Without structured dialectical prompting, models politely agree with each other.
- **Uninterruptible Playback:** Users are passive listeners locked into pre-rendered audio files.
- **Uniform Tone:** Single-voice readers lack distinct emotional prosody, gender diversity, and host presence.

✓ The AI Roundtable Solution

- **Snappy 35-50 Word Turns:** Enforces natural spoken cadence with direct name-calling ("Sophia is missing Marcus's point").
- **Dialectical Triad:** Moderator steers the flow; Optimist drives forward thesis; Skeptic stresses edge cases.
- **Live Human Interruption:** Halts streaming audio at 0ms, updates context, and forces agents to address the listener.
- **ElevenLabs Low-Latency Streaming:** Uses Turbo v2.5 to stream natural audio chunks with distinct personalities.

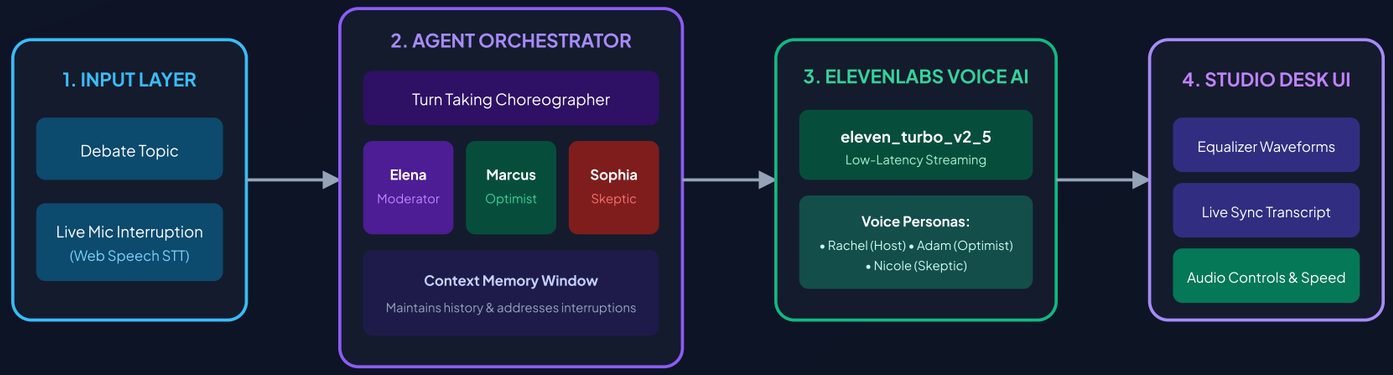
Technologies Used & Their Purpose

Engineered precisely according to hackathon voice-native specifications.

TECHNOLOGY	PURPOSE IN AI ROUNDTABLE	IMPLEMENTATION DETAILS
Python or Node.js	Backend logic and multi-agent orchestration	`Node.js Express / orchestrator.js`
LLM API	Generates each agent's arguments, counterpoints, and banter	`Gemini 2.0 Flash / OpenAI / Dynamic`
ElevenLabs Text-to-Speech	Gives each agent a distinct voice with low-latency streaming	`eleven_turbo_v2_5 (Rachel, Adam, Nicole)`
WebSockets or HTTP Streams	Supports real-time audio chunk delivery and instant event signaling	`Binary chunk stream / SSE / REST`
React / Next.js	Frontend, circular studio desk, playback controls and transcript	`React 19, Vite, Tailwind CSS`
Speech-to-Text	Converts the user's spoken questions into text in real time	`Web Speech API (webkitSpeechRecognition)`
In-Memory State / History	Tracks conversation history, active speaker queue, and whose turn it is	`Turn Memory Window & Interruption Stack`

End-to-End Voice AI Architecture

From topic initiation & human interruption to low-latency audio rendering.



Autonomous Cast & Banter Dynamics

Role-based prompting creates authentic debate without training separate models.

EV

MODERATOR

Elena Vance

`ElevenLabs: Rachel (21m00...)`

Maintains conversational pace, frames intellectual dilemmas, challenges weak claims, and gracefully transitions when the listener interrupts.

Prompt Rule: "Keep turns punchy (35–50 words). Address co-panelists by name."

MS

THE OPTIMIST

Dr. Marcus Sterling

`ElevenLabs: Adam (pNInz6...)`

Champions developer leverage, historical parallels (compilers, industrial scale), and infinite demand for intelligent software.

Prompt Rule: "Present the strongest thesis in favor; counter risks with progress data."

SC

THE SKEPTIC

Sophia Chen

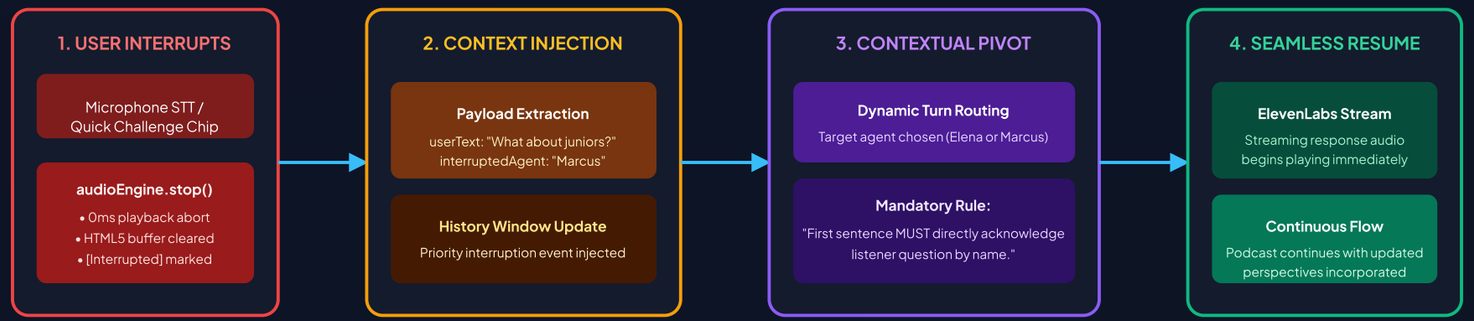
`ElevenLabs: Nicole (piTKgc...)`

Unpacks hallucination edge cases, legal liability, compute constraints, and the collapse of the junior developer apprenticeship ladder.

Prompt Rule: "Challenge ungrounded hype; insist on accountability and real-world failure modes."

Real-Time Human Interruption Lifecycle

How the system stops streaming speech in 0ms and pivots the debate without repeating itself.



Live Demo: Turn-by-Turn Script

Topic: "Will AI replace software developers by 2028?"

Elena (Host) *"Welcome to AI Roundtable. Today's question: 'Will AI replace software developers by 2028?'. Marcus, you claim this multiplies developer leverage. Start us off—what is your core thesis?"*

Marcus (Optimist) *"Thanks Elena. Throughout history, every abstraction layer created more value. Compilers didn't eliminate coders; they allowed software to eat the world. Every engineer now commands a 10× leverage team!"*

Sophia (Skeptic) *"Compilers were deterministic, Marcus. LLMs hallucinate false code. And if companies automate away junior roles, who trains the senior software architects of 2030?"*

Audience Interrupt [INTERRUPTS MARCUS] "Wait Marcus, what about the college grads who can't even get an entry-level interview today because companies expect 5 years of senior experience?"

Marcus (Pivot) *"Crucial point from our listener. While traditional junior tasks are shifting, software demand is effectively infinite. Apprenticeship will evolve into verification, system architecture, and domain modeling."*

The Future of Voice-Native AI

Shifting from passive, one-way AI speech to interactive, participatory conversational media.

Interactive Education

Students step into historical roundtables (e.g. Einstein vs Bohr) and challenge historical figures directly.

Corporate & Market Intel

Autonomous bull/bear agent debates analyzing earnings calls, regulatory filings, and market competitors.

WebRTC Full-Duplex Audio

Sub-150ms bi-directional voice streaming with acoustic echo cancellation and acoustic voice activity detection.

PROJECT AUTHORS

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